Ajeet Krishnasamy

Dallas, TX | +1-647-471-2271 | ajeetkrish@icloud.com | LinkedIn | Portfolio

Experience

Systems Engineer, Medical Equipment (Intern) — Oct 2020 – Sept 2021
CenTrak — *Newtown, PA, United States*

- Improved device reliability assessments by developing custom Python scripts and Excel models to analyze real-time medical device performance data, ensuring higher precision in monitoring systems.
- Increased maintenance planning efficiency by building automated technical performance reports for Connect Pulse™, providing engineering and operations teams with actionable insights to reduce system downtime.
- Optimized hardware lifecycle management by interpreting battery discharge and device datasets with Python (Pandas, NumPy) to identify failure trends, enabling data-driven decisions that prevented unexpected equipment outages.
- Streamlined troubleshooting workflows by structuring large-scale RTLS datasets and applying Python-based data-cleaning techniques to generate actionable findings that accelerated technical support resolutions.

Systems Engineer, Medical Equipment (Intern) — July 2019 – Sept 2019
CenTrak — *Newtown, PA, United States*

- Enhanced Real-Time Location Services (RTLS) functionality by applying Python-based regression and classification techniques to performance data, identifying system bottlenecks and refining location accuracy.
- Increased system validation accuracy by executing rigorous on-site testing protocols and coordinating cross-functional feedback loops with engineering teams to bridge the gap between lab results and field performance.
- Streamlined stakeholder decision-making processes by synthesizing complex technical datasets into actionable findings and formal recommendations using Python and Excel, ensuring data-driven alignment on system upgrades and feature rollouts.

Projects

Airfoil CFD Explorer: Automated Aerodynamic Analysis & Shape Optimization | *Python, CFD (SU2)*

- Built an automated CFD pipeline solving compressible RANS on hybrid C-grid meshes across 5 angles of attack ($0° - 16°$), generating Cl/Cd polars with convergence history for each regime.
- Validated NACA 0012 aerodynamic coefficients against Ladson 1988 wind tunnel data ($\text{Re} = 1 \times 10^6$, $\text{M} = 0.15$), benchmarking SU2 RANS predictions against experimental lift and drag polars.
- Optimized an airfoil at $\alpha = 4°$ via CST parameterization (16 Bernstein weights), NeuralFoil, and SciPy SLSQP; minimizing drag at target lift under geometric constraints, with SU2 RANS verification.

The Aerodynamics of Soccer: CFD Analysis of Spin, Wake, and Tactical Flow | *Python, CFD (SU2)*

- Built a two-solver CFD pipeline (ΦFlow laminar + SU2 RANS k-ω) solving the incompressible Navier-Stokes equations to simulate Magnus effect, vortex shedding, and wake dynamics in soccer.
- Simulated a spinning cylinder (Magnus) vs. no-spin knuckleball at $\text{Re} \approx 4 \times 10^4$, modelling real ball drag between ΦFlow Cd ≈ 1.2 (laminar) and SU2 Cd ≈ 0.68 (turbulent).
- Performed CFD wake-drafting analysis on overlapping fullbacks ($\text{Re} \approx 2 \times 10^4$): 42.2% drag reduction at 2 m inline, decaying to 27.8% at 4 m.

Wearable Hip Protector (Capstone Project) | *SolidWorks, FEA, MATLAB*

- Designed a wearable hip-protection system to mitigate impact forces from lateral and posterolateral falls in older adults.
- Conducted ANSYS finite element stress simulations on padding layers to optimize impact absorption under dynamic loading.
- Developed a MATLAB GUI that parameterizes pad sizing and strap geometry from user body measurements.
- Prototyped multilayer padding architecture using 3D printing, iterating on fit and flexibility through hands-on testing cycles, and designed an adjustable harness-style attachment for secure anatomical alignment.

Technical Skills

Data Analysis: Python (NumPy, Pandas, SciPy, Matplotlib), MATLAB, Excel
CAD & Simulation: SolidWorks, ANSYS, FreeCAD, Fusion 360, SU2
Manufacturing: CNC Machining, 3D Printing, Lathe & Mill Operations, MIG Welding

Education

University of Ottawa — Ottawa, ON, Canada
Bachelor of Applied Science (BASc), Biomedical Mechanical Engineering — *May 2026*